



POLITECNICO
MILANO 1863

SPACECRAFT ATTITUDE DYNAMICS AND CONTROL

Lab 5 – Attitude Kinematics (2)

PROF. FRANCO BERNELLI

PIETRO BRUSCHI, MASSIMILIANO BUSSOLINO, ANNACHIARA IPPOLITO, ANDREA COLAGROSSI

A.Y. 2025-2026

Task 1: Simulate the kinematics using Euler Angles

Principal moments of Inertia

$$I_x = 0.0700 \text{ kg m}^2$$

$$I_y = 0.0550 \text{ kg m}^2$$

$$I_z = 0.0250 \text{ kg m}^2$$

Initial conditions

[You choose the I.C.s]

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_x \omega_z \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_y \omega_x \end{cases}$$

Kinematics

$$\begin{cases} \dot{\phi} = \frac{\omega_z \cos(\psi) - \omega_x \sin(\psi)}{\cos(\vartheta)} \\ \dot{\vartheta} = \omega_x \cos(\psi) + \omega_z \sin(\psi) \\ \dot{\psi} = \omega_y - [\omega_z \cos(\psi) - \omega_x \sin(\psi)] \frac{\sin(\vartheta)}{\cos(\vartheta)} \end{cases}$$

Tasks

- Simulate the kinematics by accounting for the presence of singularities
- Build the DCM from the Euler Angles

$$A_{312} = \begin{bmatrix} \cos \psi \cos \phi - \sin \psi \sin \phi \sin \vartheta & \cos \psi \sin \phi + \sin \psi \cos \phi \sin \vartheta & -\sin \psi \cos \vartheta \\ -\sin \phi \cos \vartheta & \cos \phi \cos \vartheta & \sin \vartheta \\ \sin \psi \cos \phi + \cos \psi \sin \phi \sin \vartheta & \sin \psi \sin \phi - \cos \psi \cos \phi \sin \vartheta & \cos \vartheta \cos \psi \end{bmatrix}$$

Task 2: Simulate the pointing error of a spin-stabilized S/C

Principal moments of Inertia

$$I_x = 0.0700 \text{ kg m}^2$$

$$I_y = 0.0550 \text{ kg m}^2$$

$$I_z = 0.0250 \text{ kg m}^2$$

Initial conditions

$$\omega_x(0) = \pm C \text{ rad/s}$$

$$\omega_y(0) = 0.1 \text{ rad/s}$$

$$\omega_z(0) = 0.1 \text{ rad/s}$$

$$C \in [0.2, 2\pi] \text{ rad/s}$$

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_x \omega_z \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_y \omega_x \end{cases}$$

Kinematics

$$\begin{cases} \dot{\phi} = \frac{\omega_z \cos(\psi) - \omega_x \sin(\psi)}{\cos(\vartheta)} \\ \dot{\vartheta} = \omega_x \cos(\psi) + \omega_z \sin(\psi) \\ \dot{\psi} = \omega_y - [\omega_z \cos(\psi) - \omega_x \sin(\psi)] \frac{\sin(\vartheta)}{\cos(\vartheta)} \end{cases}$$

Compute initial attitude parameters such that, in inertial frame, $\mathbf{h}(0) \parallel [1, 0, 0]^T$

Tasks

- Simulate the kinematics by accounting for the presence of singularities
- Compute the pointing error with respect to the nominal direction $[1, 0, 0]^T$ in the inertial frame